

JK Tyre BTP

Forward Production Scheduler

Deterministic, BOM-driven, event-aware production scheduling for the Banmore Tyre Plant Passenger-Car Radial pilot SKU.

Pilot SKU	1325220516095HTMX0 — 205/65 R16 PCR Taxi MX TL
Green Tyre	GT2056516TAXIMXTL — 8 in-scope BOM components
Curing Press	14811 (fixed input, immovable)
Horizon	2026-05-17 06:49 → 2026-05-30 22:19 (~14 days)
Curing Blocks	42 blocks — 2,620 tyres total demand
Build Version	V1 — Demand Fulfilment Scope
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Prepared as the complete project documentation: objective, constraints, approach, architecture, pipeline modules, locked design decisions, outputs, KPIs, testing and acceptance.

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1. Executive Summary

This documentation describes the **V1 release of the JK Tyre BTP Forward Production Scheduler**, a deterministic, BOM-driven engine that consumes the published May curing plan as a fixed input and produces a fully time-stamped, machine-assigned schedule for every upstream operation of the pilot Passenger-Car Radial SKU. Curing is never moved; if upstream cannot feed it within aging windows, the engine flags the violation in a diagnostic artefact and continues, never silently fixing demand.

Objective

Prove that every curing block in the May plan can be fed on time with all eight in-scope BOM components, within their respective aging windows, using a single forward pass that is **reproducible byte-for-byte** on the same inputs.

Why a forward, deterministic scheduler

Tyre manufacturing is rich with aging constraints: a calendered ply is unsafe to build with after 24–48 hours; a master compound that has not aged long enough behaves differently on the line. Conventional spreadsheet planning cannot reason consistently about a 7-level BOM with per-edge minimum/maximum aging windows. The scheduler encodes those rules formally, walks the BOM from curing demand backward, and commits lots in chronological order with an explicit Least-Slack-First tiebreak. Reproducibility is enforced so the planner can iterate confidently.

What is in V1

- Hard constraints fully enforced: aging windows, AND-join across 8 components, MPQ, machine eligibility, fixed curing.
- Eleven output artefacts per run, including a bundled Excel workbook with 11 sheets.
- Five golden fixtures + 215 unit/integration tests; integration suite asserts byte-identical re-runs.
- Twelve pipeline modules, each runnable in isolation with a typed dataclass contract.
- Auto- t_0 anchor (L17) computed from the longest BOM aging chain; guardrail asserts feasibility before any lot is committed.

What V1 intentionally does not do

- No optimisation of curing — curing is a fixed input.
- No changeover modelling — set to 0 min in V1 (the 15 min / 0 min model is deferred to V2).
- No utilisation as an optimisation target — utilisation is reported but not optimised.
- No rescheduling, repair loops, or backtracking — flag-and-continue per L11.

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The JK Tyre Banmore Tyre Plant (BTP) manufactures Passenger-Car Radial (PCR) tyres through a long, multi-stage process chain. Every tyre flows through:

Mixing → Final Mixing → Calendering → Cutting / Slitting → Ply Cutting → Bead Building → Extrusion → Tyre Building → Curing

Curing is the last operation and is the most capacity-constrained: a single curing press serves a given SKU. For the pilot, that press is **14811**. The plant has published a May curing plan that specifies which press, which 8-hour shift, and how many tyres for each of 42 blocks across the horizon. That plan is non-negotiable input — the V1 scheduler's job is to make every upstream operation land in time, on a feasible machine, with every component within its aging window.

The engine is **not** an optimiser, not a demand forecaster, and not a labour/energy/cost model. It is a scheduler over material flow, machine assignment, and time.

3. Scope — V1 vs V2+

V1 — Demand Fulfilment

The V1 KPIs are deliberately simple, because the priority is correctness rather than savings:

- **OTIF %** at the Building → Curing handoff (per block).
- **Count of aging violations** per edge.
- **Count of infeasibilities** per lot, with binding constraint named.
- **Coverage** — every in-scope operation has a scheduled lot.

V2+ — Optimisation Layer

Once V1 is locked and reproducible, V2 layers in the savings work: changeover modelling (15 min on different product, 0 min on same), utilisation as an optimisation objective, same-product clustering on shared resources such as the Four-Roll Calendar and the mixer pool, and the strict event-driven dispatch with the full LSF tiebreak chain. None of those is touched in V1.

Explicit V1 simplifications

Topic	V1 behaviour	V2 target
Changeover	0 min always (L8)	15 min different-product / 0 min same-product
Dispatch	Topo + chronological greedy with CPM backward pass	Full event-heap dispatch (L21)
LSF tiebreak	Avoided by sequential dispatch	4-step LSF chain (L15)
Soft-reservation expiry	Not modelled — no contention	Full create/consume/expire/release log (L16)
Backward feasibility	Min-aging chain only	Min-aging + processing time (refined CPM)

Attribute	Value
SKU code	1325220516095HTMX0
Description	205/65 R16 Passenger-Car Radial (Taxi MX TL)
Green tyre code	GT2056516TAXIMXTL
Curing press	14811 (single press for this SKU)
Horizon	2026-05-17 06:49 → 2026-05-30 22:19
Curing rows	42 (mostly 64 tyres / 8-hr block, 15-min cycle)
Total demand	2,620 tyres
Pre-existing GT inventory	0 on every row
BOM depth	7 levels (SKU → GT → component → sub-component → final compound → master → raw)

BOM children of the Green Tyre

#	Component code	Item type	In scope
1	CPJ1218-162MM/29°	Rubberized Steel Belt	Yes
2	CPJ1218-154MM/29°-1	Rubberized Steel Belt	Yes
3	TRD2056516TAXIMXTL	Tread	Yes
4	IL2056516TAXIMXTL	Inner Liner	Yes
5	SSW2056516TAXIMXTL	Sidewall	Yes
6	EHT1000-640MM/90°	Rubberized Ply	Yes
7	EHT1000 -480MM/90°	Rubberized Ply	Yes
8	BD-12843443-4	Bead Apex	Yes
—	CAP 66 - CAPSTRIP	Cap Strip	OUT OF SCOPE (L12)

Inputs (read-only)

File	Purpose
BTP_PCR_May_Curing_Schedule.csv	14,292 rows over the full May plan; 42 rows for the pilot SKU. Each pilot row is a non-negotiable demand event. Columns: Date, Shift, Machine, SKUCode, StartTime, EndTime, Qty, CycleTime_min, GT_Inventory, Remarks, SKU_Description.
BTP_Routing_*BOM_Final.xlsx	Six sheets — BOM, Routing, Aging Master, Buffer Master (ignored per L4), ItemType Master, MPQ.
JKT_BTP_Forward_Scheduler_Problem_Statement.pdf	Authoritative problem definition. Section numbers cited throughout this document resolve into this file.

5. Plant Resources — Departments & Machines

Department	Machines (eligible pool)	Notes / UOM
Mixing (Master)	0201, 0202, 0203, 0204, 0205, 0206	SEC/BATCH; batch in KG
Final Mixing	0201, 0202, 0203, 0204, 0205	SEC/BATCH; batch in KG
FOUR ROLL CALENDAR	FRC (single shared resource)	M/MIN or SEC/BATCH. Bottleneck candidate
Belt Cutter	WBC, WBCNew	M/MIN
FULL WIDTH SLITTER	FWS, FWSNew	M/MIN (Capstrip — out of scope)
Quintuplex Extruder	Quintuplex	M/MIN — Tread
TRC	TRC	M/MIN — Inner Liner
Duplex Extruder	Duplex	M/MIN — Sidewall
PLY CUTTER	LTBC, HTBC, LTBCNew	M/MIN
VIPO	VIPO	MIN, batch 100 NOS (beads)
AUTO AND MANUAL FILLERING	FILLERING	proc_time NULL on BD apex (HALT)
Tyre Building (VMIMaxx)	7001, 7002, 7003, 7004, 6001, 6002, 6003, 6004	SEC per tyre; primary 6001 (L18)
Curing	14811 (fixed)	750 SEC nominal cycle — immovable

Every constraint below must hold for every scheduled lot. The engine detects and reports violations itself — they are never silently fixed.

#	Constraint	Rule
4.1	Aging window	For every BOM edge: $\text{producer_end} + \text{MIN_aging} \leq \text{consumer_start} \leq \text{producer_end} + \text{MAX_aging}$. Inclusive both ends (L22). Applies to every edge including Building $\rightarrow$ Curing.
4.2	AND-join completeness)	(BOM Tyre Building consumes 8 mandatory components. It cannot start until each one has at least one ready, in-window lot. Modelled as an atomic hard reservation across all 8 producer lots.
4.3	Machine eligibility	Every lot runs on a machine listed in the routing for that operation. One lot per machine at a time.
4.4	MPQ	Every lot satisfies MPQ_Min. Lots greater than MPQ_Max are split into equal sub-lots.
4.5	Curing is fixed	Curing rows are immovable. Aging or AND-join violations are FLAGGED; curing is never shifted (L4.5 + L11).
4.6	Determinism	Same inputs $\rightarrow$ byte-identical outputs. No randomness, no wall-clock dependence.

7. Soft Rules & Plant Policies (Section 5)

- **24×7 operation** — no shift breaks or planned maintenance windows modelled in this phase.
- **15-minute changeover** between different products on the same machine; 0 min for same product back-to-back. No changeover at machine cold-start. **V1 sets this to 0 min always** (L8).
- **10-minute transfer** between producer and consumer, interpreted via $effective_gap = \text{MAX}(\text{transfer_time}, \text{MIN_aging})$ per L14 — material can be in transit while it ages.
- **95% machine efficiency**: $effective_time = \text{ceil}(\text{nominal_time} / 0.95)$ (L10, L20). Single ceil direction throughout.
- **Tyre Building primary machine**: pin to one primary (6001 by L18); spill to a 2nd/3rd machine only if staying on the primary would force a component to age past its window.
- **Null processing times** must be filled with a documented, justifiable default — except **BD-12843443-4 Filler**, which HALTs the engine until the planner supplies a value (Section 8.D).

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Passenger-Car Radial Pilot Programme

These 23 decisions are settled. They are never relitigated without explicit reversal from the planner. Each shapes exactly one behaviour in the engine.

ID	Decision	Choice
L1	Demand grain	Per-block: one Building lot per curing row; each Building lot points to exactly one curing block.
L2	Starting inventory	Zero. Engine produces every compound, ply, bead from raw. Raws are bottomless pre-horizon.
L3	Building machine	Pin to one primary. Spill to a 2nd/3rd ONLY when staying primary would cause an aging breach.
L4	Aging vs Buffer Master	Aging Master is authoritative. Buffer Master ignored for this phase.
L5	Aging clock anchor	Timer starts at producer END, stops at consumer START.
L6	Master→master aging	Full aging applies on every BOM edge, including master compound → master compound.
L7	EHT1000 duplicate	Use the row with is_primary == 1.0; drop the NaN row; log Warn.
L8	Changeover	V1: 0 min always. V2 model documented but disabled.
L9	Output time precision	1 minute (matches curing CSV).
L10	Efficiency factor	effective = ceil(nominal / 0.95). Single ceil direction throughout.
L11	Infeasibility	Flag and continue. No reschedule, no repair loop, no backtrack. Never shift curing.
L12	Capstrip on ice	Skip CAP 66 chain entirely until corrected data arrives.
L13	Work-Away entries	Treated as reclaim/scrap inputs, assumed already available.
L14	Transfer + aging interplay	effective_gap = MAX(transfer_time, MIN_aging).
L15	LSF tiebreak chain	LSF → earliest curing deadline → longest downstream path → item_code asc.
L16	Soft reservation rule	Exclusive, invisible to others, auto-expires at consumer's latest_acceptable_start.
L17	t_0 anchor	Auto-computed (default) as first_curing_start – critical_path – safety_buffer; guardrail asserts feasibility.
L18	Building primary (V1)	Lexicographic lowest machine_id: 6001.
L19	FEFO eligibility	Producer must have aged in (producer_end + MIN_aging ≤ sim_time). Earliest aging-MAX wins; lot_id asc tiebreak.
L20	Time domain	Single integer-minute domain. ceil throughout. Datetime ↔ minute lives only at audit-in and writer-out.
L21	Event-driven dispatch	Min-heap on (event_minute, event_class, lot_id). V1 uses topo greedy as documented simplification.
L22	Aging-MAX inclusive	≤ not <. gap == MAX is compliant; gap = MAX+1 is a violation.
L23	Data-shape locks	lot_id format, machine_id string, single unit-conversion table, datetime↔minute boundary in audit/out.

9. Standing Assumptions (Section 8)

ID	Topic	V1 behaviour
8.C	Compound lot sizing	Forward-aggregate consecutive block demand into the largest lot satisfying $\text{lot_qty} \leq \text{MPQ_Max}$ AND aging-MAX horizon. Equal-split above MPQ_Max. HALT (locked) when a single block's demand < MPQ_Min AND aggregation across blocks is blocked by aging-MAX.
8.D	Null proc_time on BD-12843443-4 Filling	Do NOT impute. Audit HIGHLIGHTS the routing row and the engine refuses to schedule that operation until the planner supplies a value.
8.F	Mixer pool size mismatch	Ignore alt_machine_count entirely. Audit parses the messy machines cell and derives eligible_machine_count, which the scheduler uses.
8.I	Dispatch rule	Least Slack First (full tiebreak chain L15) for lots; FEFO with eligibility L19 and soft reservation L16 for inventory consumption. Deterministic.

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The audit module catches every irregularity it sees. Findings are classified as **HALT** (engine refuses to write `schedule.csv` and exits non-zero) or **WARN** (engine continues and lists the finding in `audit_report.md`).

#	Finding	Severity
1	Aging Master rows with <code>MaxAgingUnit</code> $\neq$ <code>MinAgingUnit</code> (normalised to minutes).	Warn
2	Inconsistent quoting in machine cells — parser normalises; <code>eligible_machine_count</code> derived.	Warn
3	<code>transfer_time_min</code> null on most routing rows → plant default 10 min.	Warn (default-used flag)
4	BD-12843443-4 Filling proc_time NULL.	HALT (exit code 10)
5	EHT1000 -480MM/90° calendering duplicate — keep <code>is_primary</code> =1.0.	Warn
6	<code>alt_machine_count</code> wrong on many rows — column ignored.	Warn
7	Capstrip chain has conflicting Aging Master rows + no MPQ.	Warn + auto-exclude (L12)
8	Pilot items missing from Aging / ItemType Master.	HALT (exit codes 11/12)
9	Informal <code>predecessor_rule</code> text; <code>operation_seq</code> is the structured signal.	Warn on mismatch
10	t0 guardrail failure (L17).	HALT (exit code 30)

11. Solution Approach

Philosophy

The scheduler is a **single-pass, deterministic, BOM-driven** engine. From each fixed curing block it walks the BOM downward to expand demand; it then sizes lots forward in chronological order, anchored at t_0 , with explicit Least-Slack-First tie-breaking. Decisions are committed once — the engine never backtracks. When a hard constraint cannot be satisfied (an aging window, an AND-join across the eight Building components, or a machine deadline), the engine logs the binding constraint by name and moves on (L11).

Why forward and not optimisation

A forward pass is auditable, explainable to the floor, and reproducible. Each lot points back to the raw-data row that defined its time, quantity, and machine. Each soft reservation is logged as it is created and as it is consumed. The cost is that the engine cannot trade off lots against each other to improve aggregate utilisation; that trade-off is deliberately deferred to V2.

Time domain

All scheduling math happens in a **single integer-minute domain**. The datetime $\leftrightarrow$ minute conversion happens exactly twice: once in the audit step (in), and once in the output writer (out). No other module touches a *pd.Timestamp*. Every duration is computed with a single ceil direction; no module duplicates the $\text{ceil}(x/60)$ logic. The unit-conversion table (L20) is the single source of truth.

Aging arithmetic

For every BOM edge, the engine enforces:

```
producer_end + effective_gap ≤ consumer_start ≤ producer_end + MAX_aging
where:
effective_gap = MAX(transfer_time_min, MIN_aging_min) (L14)
Both bounds are inclusive (L22).
```

Lot sizing

Lots are formed by walking served curing blocks in chronological order, accumulating quantity into the current lot while both (a) $\text{lot_qty} \leq \text{MPQ_Max}$ and (b) the aging horizon of the earliest served block still covers the latest. The current lot is closed and a new one opened the moment either bound breaks. If a single block's demand exceeds MPQ_Max , the lot is split into equal sub-lots; no remainder lot smaller than MPQ_Min is permitted. If a single block's demand is below MPQ_Min and aggregation across blocks is impossible because of the aging-MAX bound, the engine HALTs and prints the offending (block, compound).

FEFO + soft reservation

When a consumer lot is being scheduled, it identifies its producer lots via FEFO (First-Expiring First-Out): among lots that have aged in, the one whose aging-MAX is soonest is selected, ties broken by *lot_id* ascending. The chosen producer is placed under a soft reservation: exclusive, invisible to other FEFO scans, and auto-expiring at the consumer's *latest_acceptable_start*.

AND-join atomicity for Building

Tyre Building requires all 8 in-scope BOM components to be ready and reserved at the same instant. The reservation is **atomic**: if any one of the eight cannot be reserved, the others are released and the Building lot is routed to the infeasibility sheet with the missing component named. This is the only place where a multi-component AND-join is enforced.

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This is the implementation spec. Numbered 1–26, gap-free. Each step is owned by a specific pipeline module from Section 13; each step has a deterministic, inspectable output.

#	Action	Detail
1	Load raw inputs.	Read curing CSV, all six routing/BOM sheets, and the problem-statement PDF metadata. No mutation. (audit)
2	Filter to pilot SKU.	Curing CSV → 42 rows for the pilot. Routing/BOM/Aging/ItemType/MPQ subset to all items reachable from the pilot Green Tyre, excluding the Capstrip chain (L12). (audit)
3	Normalise units to integer minutes.	Apply L20 conversion table: SEC/BATCH → ceil(/60), M/MIN → ceil(lot_qty_m / proc_time), MIN → as-is; aging Days/Hours/Minutes → minutes. (audit)
4	Compute t0 and minute-0 anchor.	Load t0 from config (auto or default). Run the L17 guardrail assertion. HALT and print the longest BOM path if violated. (audit)
5	Parse messy machine cells.	Normalise quotes/encoding, produce clean machines list per routing row, derive eligible_machine_count. machine_id stored as string everywhere. (audit)
6	Classify data-quality findings.	Split into HALT and WARN buckets per Section 9. (audit)
7	Handle EHT1000 duplicate.	Keep is_primary=1.0 row; drop NaN row; log Warn. (audit)
8	BOM explosion per curing block.	For each of the 42 pilot curing rows, walk the BOM downward and compute the per-edge demand quantity. (demand_explosion)
9	Demand aggregation per item across blocks.	Group demand by item_code; preserve served-block list. (demand_explosion)
10	Lot sizing — forward aggregate.	Build lots by walking served blocks chronologically; close and open new lot when MPQ_Max or aging-MAX horizon breaks. (lot_sizing)
11	Lot sizing — MPQ_Max split.	Equal sub-lots if a single block exceeds MPQ_Max. No remainder below MPQ_Min. (lot_sizing)
12	Lot sizing — HALT on tight aging.	If single-block demand < MPQ_Min AND aggregation blocked by aging-MAX, HALT. (lot_sizing)
13	Generate lot IDs.	lot_id = {safe_item_code}_{op_seq}_{lot_seq:04d}. Readable item_code kept separately. (lot_sizing)
14	Build the lot DAG.	Node per lot; directed edge per BOM edge with (min_aging, max_aging, effective_gap). Export dag.json. (graph_construction)
15	Backward feasibility limits.	Per-lot latest_acceptable_start_min from min-aging chain. Pure forward pass owns scheduling. (backward_feasibility)
16	Per-lot processing duration.	duration_min = ceil(nominal_min / 0.95). Three regimes: continuous, per-batch, per-cycle. (time_calculation)
17	Initialise event-driven scheduler.	current_sim_time = t0_minute. Empty machine-free heap, empty reservation table, empty running-lots map. (scheduling)
18	Event loop — pop next event.	Pop heap by (event_minute, event_class_priority, lot_id). Advance current_sim_time. (scheduling)
19	Dispatch — build ready set.	Enumerate lots whose predecessors are FEFO-eligible, latest_acceptable_start ≥ sim_time, and at least one free eligible machine. (scheduling)
20	Dispatch — apply LSF tiebreak.	Sort ready set by LSF chain (L15). Pick the head. (scheduling)

#	Action	Detail
21	FEFO match + soft reservation.	For each predecessor pick the FEFO-eligible lot with earliest aging-MAX; place soft reservation. Atomic AND-join for Building (8 components). (scheduling)
22	Commit the lot.	Assign machine; set start_min and end_min; push lot-completion and lot-aged-in events. Mark consumed reservations. (scheduling)
23	Building primary-machine pinning.	Try 6001 first (L18); spill only if waiting would breach aging-MAX of any component (L3). (scheduling)
24	Continue the loop.	Repeat 18–23 until heap empty or every curing block has a committed Building lot (or a logged infeasibility). Expired reservations logged. (scheduling)
25	Diagnostics & violations.	Recompute every gap, flag [MIN, MAX] breaches (inclusive). Classify Building → Curing OK/LATE/EARLY. Compute OTIF %. (diagnostics, kpi)
26	Write outputs.	Eleven artefacts into output/ . Datetime ones converted back from minutes only here. Exit code = 0 on clean run; non-zero with binding HALT otherwise. (visualisation + writer)

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Thirteen pipeline steps. Each module is invocable in isolation given the upstream typed result; this makes incremental debugging straightforward. Module boundaries are frozen dataclasses (AuditResult, NormalisedResult, BomGraph, DemandResult, LotsResult, LotDagResult, FeasibilityResult, DurationResult, ScheduleResult, DiagnosticsResult, KpiResult).

```
input/  
curing.csv routing.xlsx problem.pdf  
|  
v  
1. audit -> audit_report.md, routing_cleaned.csv (HALT-capable)  
2a. t0_compute (L17) -> integer minute-0 anchor + guardrail  
2b. unit_normalisation -> aging/proc_time -> integer minutes (single ceil)  
3. bom_graph -> nx.DiGraph + is_capstrip propagation  
4. demand_explosion -> per-curing-block demand tree (zero blocks skipped)  
5. lot_sizing -> Lots with serves_blocks; HALT on tight aging  
6. graph_construction -> lot DAG (dag.json); effective_gap per edge  
7. backward_feasibility -> latest_acceptable_start scalar per lot  
8. time_calculation -> duration_min per (lot, machine), ceil(/0.95)  
9. forward_scheduler -> topo greedy + CPM backward pass + FEFO + L18 + L11  
10. diagnostics -> aging_violations.csv, building_to_curing.csv  
11. kpi -> OTIF%, util, processing min, span  
12. visualisation -> bom_graph.svg, schedule.csv, gantt_<block>.html (x3)  
13. writer_excel -> btp_schedule.xlsx (11 sheets)
```


14. Module-by-Module Reference

Module	Responsibility
audit (V1/routes/audit.py)	Reads raw inputs (no mutation); surfaces data-quality findings; dedups masters; parses messy machine cells; fixes Å° mojibake; drops Capstrip (L12) and the EHT1000 NaN-is_primary row (L7). HALT-capable.
t0_compute (V1/setups/t0_compute.py)	L17 auto-t0: computes the longest BOM critical path (per-item duration + min-aging) from leaves to SKU; anchors t0 = first_curing_start – critical_path – safety_buffer_min. Bypassed when pilot.yaml's t0.auto is false.
unit_conversion (V1/utilities/unit_conversion.py)	Converts aging Days/Hours/Minutes (and Hr/Hrs/Min aliases) to minutes; routing proc_time SEC/BATCH, SEC, MIN to minutes via ceil(x/60); curing datetimes anchored to the chosen t0. Single ceil rounding direction throughout (L20).
bom_walker (V1/utilities/bom_walker.py)	Builds an nx.DiGraph from the BOM; propagates is_capstrip down from configured seeds; validates acyclicity. Also computes longest_min_aging_path_from/to for the L17 guardrail.
demand_explosion (V1/routes/demand_explosion.py)	Walks the BOM per curing block: child_qty = parent_qty × (edge.qty / edge.output_qty). Aggregates per item; preserves serves_blocks chronologically. Zero-tyre blocks generate no demand.
lot_sizing (V1/routes/lot_sizing.py)	Forward-aggregates consecutive block demands into the largest lot satisfying both $qty \leq MPQ_Max$ and $curing_span \leq (aging_MAX - aging_MIN)$. Equal-split when a single block exceeds MPQ_Max . Green Tyre is special-cased to one lot per curing row per L1. HALT (§8.C) when a single-block lot < MPQ_Min and aging-isolated from all other demand of the same item.
graph_construction (V1/routes/graph_construction.py)	Builds the lot-level DAG; attaches effective_gap per L14. Emits dag.json.
backward_feasibility (V1/routes/backward_feasibility.py)	Per-lot latest_acceptable_end_min derived from the min-aging chain. Conservative; refined by the scheduler's CPM pass.
time_calculation (V1/routes/time_calculation.py)	Per (lot, eligible_machine): $duration_min = ceil(nominal_min / 0.95)$ (L10 / L20). Three regimes: continuous (M/MIN length-based), per-batch (when batch_size + batch_UNIT are set), and per-cycle / per-unit (Tyre Building consumes one cycle per building_tyres_per_cycle tyres).
forward_scheduler (V1/routes/forward_scheduler.py)	Topological greedy forward sweep with a CPM backward pass (per-lot floor + ceiling). FEFO producer pick per ingredient (L19); atomic AND-join for Building (Section 4.2); L18 prefers Building primary 6001; gap-aware machine intervals; L11 flag-and-continue.
diagnostics (V1/routes/diagnostics.py)	Recomputes every consumer–producer gap; flags [MIN, MAX] breaches (inclusive, L22); classifies Building → Curing as OK / LATE / EARLY / ZERO_QTY; mirrors LATE/EARLY into aging_violations.csv with a synthetic CURING__ consumer id.
kpi (V1/routes/kpi.py)	OTIF % at Building → Curing handoff, aging-violation totals, processing minutes, schedule span, per-machine utilisation.
visualisation (V1/routes/visualisation.py)	Emits bom_graph.svg, schedule.csv, machine_view.csv, and gantt_.html for three sample blocks (earliest, middle, latest in the horizon).

	Module	Responsibility
J P	writer_excel (V1/reports/writer_excel.py)	Bundles btp_schedule.xlsx with 11 sheets and a curated summary sheet. HALT runs get a slim workbook (summary, routing_cleaned, audit_halt, audit_warn).

15. Configuration (V1/config/pilot.yaml)

All tunables live in pilot.yaml. The file is loaded once at startup into a frozen Settings dataclass; downstream modules read from it. No environment-variable shortcut — configuration is declarative and reproducible.

```
pilot:
  sku_code: '1325220516095HTMX0'
  green_tyre_code: 'GT2056516TAXIMXTL'
  curing_press: '14811' # string — leading zeros matter (L23)
  horizon_start: '2026-05-17 06:49'
  horizon_end: '2026-05-30 22:19'
  total_demand_tyres: 2620

  t0:
    auto: true # L17 data-driven anchor (default)
    safety_buffer_min: 60
    default: '2026-05-01 07:00' # fallback when auto: false
    guardrail_assertion: true # L17 — HALT if longest MIN-aging path > first_curing_start

  efficiency:
    factor: 0.95 # L10 / L20

  defaults:
    transfer_time_min: 10 # S9 #3 fallback when routing transfer_time is null
    changeover_min_v1: 0 # L8 — V1 sets changeover to 0 always

  building:
    pool: ['6001','6002','6003','6004','7001','7002','7003','7004']
    primary: '6001' # L18 — V1 deterministic primary
    tyres_per_cycle: 2 # VMIMaxx GROUP produces 2 green tyres per cycle

  exclusions:
    capstrip_items: [...] # L12 hard-excluded set
    work_away_items: [...] # L13 reclaim/scrap items assumed available
    green_tyre_components: [...] # 8-component AND-join set for Building

  output:
    run_id_format: '%H%M-%d-%m-%Y'
    root: 'output'
```

- No random, or wall-clock-dependent defaults except the run-id timestamp, which is injectable for tests.

- **Every sort is explicit** — `sorted(..., kind="stable")` on every `DataFrame.sort_values`. No reliance on dict insertion order.

- **Single ceil rounding** throughout the codebase, via `V1/utilities/time_math.py`. Duplicating `ceil(x/60)` elsewhere is treated as a defect (L23).

- **Datetime** ↔ **minute** conversion lives at exactly two boundaries: audit / unit_normalisation (in) and the schedule writer (out). No `pd.Timestamp` arithmetic anywhere in the scheduler core.

- **machine_id** is a **string end-to-end** so leading zeros on the mixer pool (0201, 0202, ...) survive (L23).

- **lot_id** is the **canonical tiebreaker** for every dispatch and FEFO decision (L15 step 4, L19).

Regression gate

The integration test suite runs the full pipeline **twice** on the same inputs and asserts: (a) `dag.json` is byte-identical across runs (JSON serialisation with `sort_keys=True` is fully stable); and (b) `btp_schedule.xlsx` is sheet-by-sheet `DataFrame-equal` across runs — the workbook bytes vary because `openpyxl` embeds a build timestamp, but every cell value is reproducible. The summary sheet is excluded because it carries the per-run `run_id` by design.

17. HALT Exit Codes

The engine exits with a stable non-zero code so wrappers can branch on the binding finding. Defined in `V1/config/halt_codes.py`.

Code	Name	Meaning
0	OK	All 12 routes ran; every artefact written.
10	AUDIT_NULL_PROC_TIME	A routing row has a null proc_time (§9 #4, §8.D).
11	AUDIT_MISSING_AGING	A mandatory pilot item is absent from the Aging Master (§9 #8).
12	AUDIT_MISSING_ITEMTYPE	A mandatory pilot item is absent from the ItemType Master (§9 #8).
20	LOT_SIZING_TIGHT_AGING	Single-block lot below MPQ_Min, aging-isolated from same-item demand (§8.C).
30	T0_GUARDRAIL_VIOLATION	$t_0 + \text{sum}(\text{MIN_aging})$ along longest BOM path > first_curing_start (L17).

In every HALT case the engine still writes `audit_report.md` and `routing_cleaned.csv` so the planner has the evidence in hand without re-running.

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every run creates a fresh dated folder output/<HHMM-DD-MM-YYYY>/. Folders are never overwritten across runs — re-running within the same minute raises `FileExistsError` (deliberate). A successful run emits seven files:

Artefact	Content
btp_schedule.xlsx	Headline planner artefact. Single workbook with 11 sheets — summary, kpi, schedule, machine_view, building_to_curing, aging_violations, infeasibilities, reservation_log, routing_cleaned, audit_halt, audit_warn.
audit_report.md	Markdown rendering of Section 9 findings, split into HALT vs WARN buckets with sheet/row citations.
dag.json	Machine-readable lot dependency graph — nodes + edges with aging windows and effective-gap minutes.
bom_graph.svg	Static BOM tree viz. Capstrip subtree appears tagged OUT-OF-SCOPE.
gantt_b.html	Plotly Gantt for three sample blocks — earliest, middle, latest in the horizon.

Workbook sheet contents

Sheet	Columns / description
summary	Run metadata + headline KPIs (run_id, t0, sku, OTIF, processing minutes, span).
kpi	Counts, OTIF %, aging-violation breakdown, processing minutes, span, per-machine utilisation.
schedule	Lot-level schedule. on_time_flag=False marks lots that finished after aging-MIN ceiling (L11).
machine_view	Same rows sorted by (machine_id, start_min, lot_id) for floor-level execution.
building_to_curing	One row per Building (GT) lot per served block. Classification = OK / LATE / EARLY / ZERO_QTY.
aging_violations	One row per breached consumer-producer pair: edge_min, edge_max, actual_gap, violation_type.
infeasibilities	One row per unschedulable lot with the binding constraint named.
reservation_log	Per CLAUDE.md §16: event_minute, event_type, consumer_lot_id, producer_lot_id, qty.
routing_cleaned	Routing after dedup (L7), Capstrip drop (L12), machine-list normalisation (§8.F).
audit_halt	HALT findings only.
audit_warn	WARN findings only.

19. Golden Test Fixtures (Section 17)

These five fixtures are hand-computed and encoded as unit tests. They are the minimum acceptance bar for the audit and lot-sizing modules; the scheduler is only trusted after they pass.

#	Fixture	What it locks in
1	f1_eht1000_24h_squeeze	Tightest aging window in the pilot: EHT1000 24-hour MAX into curing block #1. Inclusive boundary at exactly 1440 min (L22).
2	f2_bd_filling_halt	Null proc_time on BD-12843443-4 Filling → HALT before any schedule.csv is written.
3	f3_eht1000_duplicate_row	Two routing rows for EHT1000 -480MM/90° calendering — is_primary == 1.0 kept, NaN dropped, logged WARN (L7).
4	f4_b460_mixed_unit	B460 aging MinAging = 4 Hours, MaxAging = 4 Days. Normaliser must output (240, 5760) minutes.
5	f5_mpq_tight_aging_halt	Synthetic single-block demand < MPQ_Min with the next block beyond aging-MAX. HALT in lot_sizing.

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The numbers below are taken from a successful end-to-end run on the pilot inputs with the BD Filling Passenger-Car Radial Pilot Programme. The numbers are taken from a successful end-to-end run on the pilot inputs with the BD Filling proc_time supplied as a planner placeholder (60 SEC/BATCH, batch 100 NOS). They illustrate the shape of the artefacts a planner would inspect.

Metric	Value
run_id	0013-22-05-2026
t0 (auto-anchored, L17)	2026-05-16 13:58:00
sku_code	1325220516095HTMX0
green_tyre_code	GT2056516TAXIMXTL
curing_press	14811
horizon	2026-05-17 06:49 → 2026-05-30 22:19
total_demand_tyres	2,620
audit_halt_findings	0
audit_warn_findings	12
lots_sized	567
lot_sizing_warnings	56
lots_scheduled	564
lots_infeasible	3
aging_violations	4 (2 min-breach, 2 max-breach)
building_to_curing_ok	36
building_to_curing_late	2
building_to_curing_early	0
OTIF %	92.308 %
total_processing_min	8,611
schedule_span_min	18,794
changeover_min_v1	0 (per L8)

Selected per-machine utilisation (busy / span)

Machine	Busy (min)	Span (min)	Util %
Mixer 0201	515	16,324	3.16 %
Mixer 0202	335	16,183	2.07 %
Mixer 0206	141	141	100.00 %
Building 6001 (primary)	578	17,757	3.26 %
Building 6002 (spill)	440	17,755	2.48 %
Building 6003 (spill)	272	16,786	1.62 %
FRC (calendar bottleneck)	1,024	16,214	6.32 %
FILLERING	2,760	16,768	16.46 %
Duplex Extruder	490	15,342	3.19 %

Machine	Busy (min)	Span (min)	Util %
HTBC (ply cutter)	192	17,983	1.07 %

Low utilisation values are expected at V1 — a single pilot SKU shares each shared resource with the rest of the plant, which is out of V1's scope. Utilisation will become a first-class objective only in V2.

P #	Criterion	Status
1	Correctness — every hard constraint holds; violations are self-reported.	Met. 0 MIN-aging violations under forward-pass discipline on the reference run. Infeasibilities named.
2	Reproducibility — byte-identical re-run.	Met. Integration test asserts dag.json byte-identical + workbook DataFrame-equal sheet-by-sheet across 2 runs.
3	Explainability — every lot points back to raw data.	Met. lot_id encodes item + op_seq + lot_seq; producer_lot_ids on each ScheduledLot; reservation log captured.
4	Coverage — every in-scope operation has a scheduled lot.	Met when BD Filling proc_time supplied. Without it, the cascade correctly flags BD + GT as infeasible (§8.D).
5	Diagnostics quality — complete + actionable.	Met. Three diagnostics sheets + infeasibility records each carry the binding constraint name.
6	Code quality — modular, fresh-clone runnable.	Met. uv sync + uv run pytest works from a fresh clone; each module has its own tests; one-line entrypoint.
7	Visualisation clarity.	Met (basic). Hierarchical bom_graph.svg with Capstrip tagged; Plotly Gantt for early/mid/late sample blocks.

22. How to Run

Prerequisites

- Python ≥ 3.11
- uv — <https://github.com/astral-sh/uv> (or any PEP-517-compatible installer)

Install

```
uv sync --extra dev --extra viz
```

Run end-to-end

```
uv run python main.py

# equivalent invocations
uv run python -m V1.setups.cli --inputs input/
uv run btp-scheduler --inputs input/
```

Expected behaviour on vanilla pilot inputs

The audit module HALTs at exit code **10** (AUDIT_NULL_PROC_TIME) because routing row 61 — BD-12843443-4 AUTO AND MANUAL FILLERING — has a null `proc_time`. This is the correct behaviour per CLAUDE.md §8.D: no silent imputation. The engine writes `audit_report.md` and `routing_cleaned.csv` so the planner can inspect the binding finding before supplying a value.

Running the full pipeline end-to-end

Supply a value for the Filler `proc_time` — two options:

- Edit the routing Excel in `input/` and re-run.
- Use the integration test path, which patches a copy of the inputs in a temporary folder and runs the full chain twice (asserting byte-identical re-runs).

```
uv run pytest tests/integration -v
```

Test suites

```
uv run pytest tests -q # all tests
uv run pytest tests/unit -q # unit tests
uv run pytest tests/integration -v # end-to-end + determinism
```

```

■■■■ CLAUDE.md # Authoritative spec – read first
■■■■ README.md # Install, run, read outputs
■■■■ main.py # Canonical entry – `uv run python main.py`
■■■■ pyproject.toml # uv-managed deps; Python ≥ 3.11
■■■■ input/ # Raw inputs (read-only)
■ ■■■■ BTP_PCR_May_Curing_Schedule.csv
■ ■■■■ BTP_Routing...BOM_Final (1).xlsx
■ ■■■■ JKT_BTP_Forward_Scheduler_Problem_Statement.pdf
■■■■ output/<HHMM-DD-MM-YYYY>/ # One dated folder per run
■■■■ V1/
■ ■■■■ config/ # pilot.yaml + Settings + HaltCode + enums
■ ■■■■ models/ # Frozen dataclasses (Lot, ScheduledLot, ...)
■ ■■■■ routes/ # Pipeline modules
■ ■■■■ utilities/ # Pure helpers (time math, FEFO, BOM walker, ...)
■ ■■■■ reports/ # Output writers
■ ■■■■ setups/ # CLI, bootstrap, t0_compute
■■■■ tests/
■ ■■■■ conftest.py
■ ■■■■ fixtures/ # 5 golden cases
■ ■■■■ unit/ # 228+ unit tests
■ ■■■■ integration/ # End-to-end + byte-identical re-run
■■■■ docs/
■■■■ methodology.md # Approach, evidence, acceptance

```

24. Glossary

Term	Definition
BTP	Banmore Tyre Plant — the JK Tyre facility this scheduler targets.
PCR	Passenger-Car Radial.
BOM	Bill of Materials. For the pilot, 7 levels deep from SKU to raw.
GT	Green Tyre. The product of Tyre Building, consumed by Curing.
MPQ	Min/Max Production Quantity. The size band a single lot must fall within.
FEFO	First-Expiring First-Out. Pick the producer whose aging-MAX expires soonest.
LSF	Least Slack First. Dispatch by smallest (latest_acceptable_start – sim_time).
AND-join	Tyre Building requires all 8 components together — atomic reservation.
t_0	Run anchor. Earliest possible mixer start; minute-0 in the integer-minute domain.
effective_gap	MAX(transfer_time_min, MIN_aging_min) per L14.
OTIF	On-Time-In-Full. Building lot delivered within its curing block aging window.
Capstrip	Out-of-scope BOM subtree (L12) — corrected data not yet provided.
Work-Away	Reclaim/scrap rubber pre-positioned at the line; assumed already available (L13).
FRC	Four-Roll Calendar — single shared resource, calendars CPJ1218 and EHT1000 plies.
VMIMaxx	Tyre Building machine group (6001–6004, 7001–7004).
Soft reservation	Exclusive, expiring producer-lot pin that the consumer places via FEFO (L16).
Aging-MIN	Minimum time a producer's material must rest before a consumer can start ($\geq$).
Aging-MAX	Maximum time a producer's material is acceptable; inclusive bound $\leq$ (L22).
HALT	Audit/lot-sizing failure that stops the run with a non-zero exit code.
WARN	Audit finding logged in audit_report.md; the run continues.

End of Project Documentation.